

# Metro Ticket Generation System

Phase 3 — UI/UX Development & Customization | ServiceNow

| Phase Purpose      | Convert the backend foundation into a usable passenger-facing Service Catalog experience.                                            |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Primary Activities | Interface design, catalog item, variables, navigation flow, Catalog Client Scripts, UI Policies and QR-ticket experience.            |
| Main Catalog Item  | Book A Metro Ticket                                                                                                                  |
| Expected Outcome   | A dynamic booking form that guides users, calculates fare values, conditionally displays fields and presents a QR-ticket experience. |

## 1. Phase 3 Overview

Phase 3 focuses on the user experience layer. The goal is to make the Metro Ticket Generation System easy to use while connecting the interface to the backend data and business logic created in Phase 2.

**Backend Data → Service Catalog Item → Variables → Dynamic Client Logic → UI Policies → QR Widget → Checkout/Ticket Experience**

## 2. UI/UX Design Principles

- Simple: Ask the passenger only for information required to generate the ticket.
- Guided: Use references, choices and conditional fields so the form explains itself.
- Dynamic: Fare values should react to station, journey type and passenger count.
- Safe: Invalid or incomplete input should be blocked before submission.
- Consistent: Use clear labels such as Starting From, Going To, Type of Journey and No. of Passengers.
- Responsive: The catalog experience should work cleanly in the intended ServiceNow interface.

## 3. Navigation Flow

**Application Navigator → Service Catalog → Maintain Items → Book A Metro Ticket → Variables / Client Scripts / UI Policies → Test Catalog Form**

## 4. Phase 3 Brainstorming

- Which fields should be visible immediately and which should appear only when needed?
- Should Starting From and Going To reference metro station records?
- Should fare fields be read-only so users cannot manually alter calculated amounts?
- What happens when passenger count changes after route selection?
- How should Single Journey and Return Journey affect displayed amount?
- When should the QR code be generated—before checkout, after confirmation, or as part of the final ticket?
- What should happen if payment mode is Others?

## 5. Activity 1 — Creation of Service Catalog Item

The primary UI entry point is a Service Catalog item named Book A Metro Ticket. It acts as the passenger booking form and hosts variables, UI policies and Catalog Client Scripts.

### Procedure

- Open the Application Navigator.
- Click All and search for Service Catalog.
- Under Service Catalog, navigate to Maintain Items.
- Click New.
- Enter the item name as Book A Metro Ticket.
- Select Catalog → Service Catalog.
- Select the Services category.
- Enter a short description if required.
- Save the catalog item.

### Suggested short description

**“Book a metro journey by selecting source, destination, journey type, passenger count and payment details. Fare is calculated dynamically.”**

## 6. Catalog Item Responsibilities

| Component              | Purpose                                                                            |
|------------------------|------------------------------------------------------------------------------------|
| Catalog Item           | Provides the user-facing booking entry point.                                      |
| Variables              | Capture journey and payment information.                                           |
| Catalog Client Scripts | Perform dynamic client-side behavior such as fare calculation and submit handling. |
| Catalog UI Policies    | Control visibility, mandatory state and behavior based on choices.                 |
| Widget                 | Provides a custom QR-ticket presentation when required.                            |

## 7. UI Structure

| Section         | Fields                                                      |
|-----------------|-------------------------------------------------------------|
| Journey Details | Starting From, Going To, Type of Journey, No. of Passengers |
| Fare Details    | Amount for Single Journey, Amount Including Return          |
| Payment         | Mode of Payment, Enter Payment Mode                         |
| Ticket Output   | QR-ticket presentation / confirmation experience            |



Reference screenshot: Service Catalog / Maintain Items navigation.

## 8. Activity 2 — Variables Configuration

Variables are the input controls displayed on the Book A Metro Ticket catalog form. Each variable should have a meaningful label, appropriate type, internal name and validation behavior.

### Variable configuration procedure

- Open the catalog item.
- Open the Variables related list and click New.
- Select reference, multiple choice, select box or single-line text as appropriate.
- Enter the variable question/label and stable internal name for scripting.
- Configure choices or reference table where required.
- Add tooltip/example text when useful.
- Set mandatory/read-only behavior according to the requirement.
- Save and test the variable on the catalog form.

## 9. Variables to Create

| S.No | Variable / Question       | Type             | Choices / Reference                    |
|------|---------------------------|------------------|----------------------------------------|
| 1    | Starting From             | Reference        | u_metro_station_detail(s)              |
| 2    | Going To                  | Reference        | u_metro_station_detail(s)              |
| 3    | Type of Journey           | Multiple Choice  | Single Journey (1), Return Journey (2) |
| 4    | No. of Passengers         | Select Box       | 1, 2, 3, 4                             |
| 5    | Amount for Single Journey | Single Line Text | Calculated fare                        |
| 6    | Amount Including Return   | Single Line Text | Calculated return fare                 |
| 7    | Mode of Payment           | Multiple Choice  | UPI, Card, Others                      |
| 8    | Enter Payment Mode        | Single Line Text | Shown when Others is selected          |

## 10. Variable Names Used by Scripts

| Display Label             | Example Internal Name     |
|---------------------------|---------------------------|
| Starting From             | starting_from             |
| Going To                  | going_to                  |
| Type of Journey           | type_of_journey           |
| No. of Passengers         | no_of_passengers          |
| Amount for Single Journey | amount_for_single_journey |
| Amount Including Return   | amount_including_return   |
| Mode of Payment           | mode_of_payment           |
| Enter Payment Mode        | enter_payment_mode        |

**Important: The exact variable names in the catalog item must match the names referenced by client scripts.**

Description

**Activity 1: Creation of Service Catalog**

- Navigate to Application navigator
- Click on All >> search for Service Catalog
- Under Service Catalog>> Maintain items
- Click on New
- Fill the details >> Name– Book A Metro Ticket
- Select Catalog>> Service Catalog
- Select Category>> Services
- Fill the Short Description, if required
- Click on Save.

Reference screenshot: variable configuration and variable list.

## 11. Catalog Client Script — Dynamic Fare Calculation

A Catalog Client Script reacts to user input. Here, the onChange script on the passenger-count variable reads source, destination and journey type, then populates the appropriate fare field.

### Configuration Procedure

- Open Service Catalog → Maintain Items.
- Open Book A Metro Ticket.
- Open the item's related lists.
- Select Catalog Client Script → New.
- Set script name and UI Type = All.
- Select Type = onChange.
- Set the variable to No. of Passengers.
- Enter the fare-calculation script and save.
- Test route, journey type and passenger-count combinations.

## 12. Fare Rules Implemented

| Route                    | Single / Passenger | Return / Passenger |
|--------------------------|--------------------|--------------------|
| Ameerpet → Madhapur      | ₹30                | ₹60                |
| Ameerpet → LB Nagar      | ₹50                | ₹100               |
| Ameerpet → Kukatpally    | ₹30                | ₹60                |
| Ameerpet → Jubilee Hills | ₹20                | ₹40                |

**Calculation rule: Total Fare = Fare Per Passenger × Number Of Passengers.**

## 13. Client Script — Core Implementation

```
function onChange(control, oldValue, newValue, isLoading) {
    if (isLoading || newValue == '') return;

    var source = g_form.getValue('starting_from');
    var destination = g_form.getValue('going_to');
    var numPassengers = Number(newValue);
    var journeyType = g_form.getValue('type_of_journey');

    if (source == 'SOURCE_ID' && destination == 'MADHAPUR_ID') {
        if (journeyType == '1') {
            g_form.setValue('amount_for_single_journey',
                numPassengers * 30);
            g_form.clearValue('amount_including_return');
        } else if (journeyType == '2') {
            g_form.clearValue('amount_for_single_journey');
            g_form.setValue('amount_including_return',
                numPassengers * 60);
        }
    }
    // Repeat the same pattern for approved routes.
    else {
        g_form.clearValue('amount_for_single_journey');
        g_form.clearValue('amount_including_return');
    }
}
```

Implementation note: Replace route placeholder IDs with the actual station sys\_ids. The supplied source also contains one extra closing brace at the end; remove it before saving.

## 14. Catalog UI Policy — Conditional Fields

A Catalog UI Policy controls presentation or behavior based on conditions. Here it is used so that Enter Payment Mode is shown only when the user selects Others for Mode of Payment.

### Scenario

**Mode of Payment = Others → Enter Payment Mode becomes visible.**

### Procedure

- Open Catalog Items and select Book A Metro Ticket.
- Open the Catalog UI Policy related list.
- Click New.
- Set Applies to = Catalog Item.
- Select Book A Metro Ticket.
- Add condition: Mode of Payment is Others.
- Save the UI Policy.
- Open UI Policy Actions and click New.
- Select Enter Payment Mode.
- Set Visible = True.
- Update and test the catalog form.

## 15. Why Use a UI Policy?

- Improves usability by hiding irrelevant fields.
- Reduces user confusion and unnecessary input.
- Centralizes simple conditional UI behavior.
- Avoids custom JavaScript when standard UI Policy behavior is sufficient.

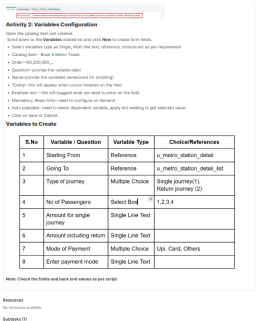
## 16. Catalog Client Script — QR Code Generation

A second Catalog Client Script is configured as an onSubmit script. It intercepts the first Checkout click, constructs a QR-code URL and opens the custom metro\_qr\_widget in a modal.

**Checkout → onSubmit → Build Ticket URL → Build QR Image URL → Open spModal → metro\_qr\_widget → Continue checkout**

### Configuration

- Open Book A Metro Ticket → Related Lists → Catalog Client Script.
- Click New.
- Set it to apply to the catalog item.
- Select Type = onSubmit.
- Enter the QR modal logic and save.
- Test that the first submission opens the QR experience instead of immediately completing checkout.



Reference screenshot: Catalog Client Script / QR-related configuration.

## 17. QR Code Client Script

```
var metroQrDone = false;

function onSubmit() {
  var sysId = g_form.getUniqueValue();
  var dummyURL = 'https://example.com/ticket?id=' + sysId;

  var qrImageUrl =
    'https://api.qrserver.com/v1/create-qr-code/?size=250x250&data=' +
    encodeURIComponent(dummyURL);

  spModal.open({
    title: "Your Metro QR Ticket – click Checkout once more to confirm",
    widget: "metro_qr_widget",
    widgetInput: { qr_url: qrImageUrl }
  }).then(function() {
    metroQrDone = true;
  });

  return false; // blocks the first Checkout click
}
```

Production note: The sample uses a placeholder ticket URL and an external QR image service. In production, the QR payload should point to a secure ticket-validation endpoint and should not expose sensitive information.

## 18. End-to-End UI/UX Flow

| Step | User Action                     | System Behavior                                                |
|------|---------------------------------|----------------------------------------------------------------|
| 1    | Open Book A Metro Ticket        | Display catalog form.                                          |
| 2    | Select Starting From / Going To | Reference variables provide valid station choices.             |
| 3    | Select Type of Journey          | Single/Return option is captured.                              |
| 4    | Select No. of Passengers        | onChange script evaluates route + journey type + count.        |
| 5    | Review fare                     | Applicable fare is populated; non-applicable field is cleared. |
| 6    | Select payment mode             | If Others is selected, Enter Payment Mode becomes visible.     |
| 7    | Click Checkout                  | onSubmit opens QR widget and blocks first submission.          |
| 8    | Confirm / Checkout again        | User continues intended ticket submission flow.                |

## 19. Phase 3 Test Cases

| Test | Scenario                            | Expected Result                                           |
|------|-------------------------------------|-----------------------------------------------------------|
| TC01 | Valid route + 1 passenger + Single  | Correct single fare populated.                            |
| TC02 | Valid route + 2 passengers + Return | Return fare = per-passenger return fare × 2.              |
| TC03 | Change passenger count              | Fare recalculates dynamically.                            |
| TC04 | Unsupported route                   | Fare fields are cleared.                                  |
| TC05 | Payment = Others                    | Enter Payment Mode becomes visible.                       |
| TC06 | Payment = UPI/Card                  | Enter Payment Mode remains hidden.                        |
| TC07 | First Checkout                      | QR modal opens and submission is blocked.                 |
| TC08 | QR confirmation                     | User can continue checkout.                               |
| TC09 | Missing journey input               | Submission/processing is prevented per validation design. |
| TC10 | Intended UI types                   | Catalog behavior is verified in the target interface(s).  |

## 20. Phase 3 Exit Criteria

- Catalog item is available and correctly categorized.
- All approved variables are created with correct types and internal names.
- Fare calculation works for every approved route/journey combination.
- UI Policy correctly controls Enter Payment Mode visibility.
- QR widget/modal behavior works as designed.
- Positive, negative and boundary test cases pass.
- Catalog experience is ready for deeper business logic, automation, notifications and production hardening.

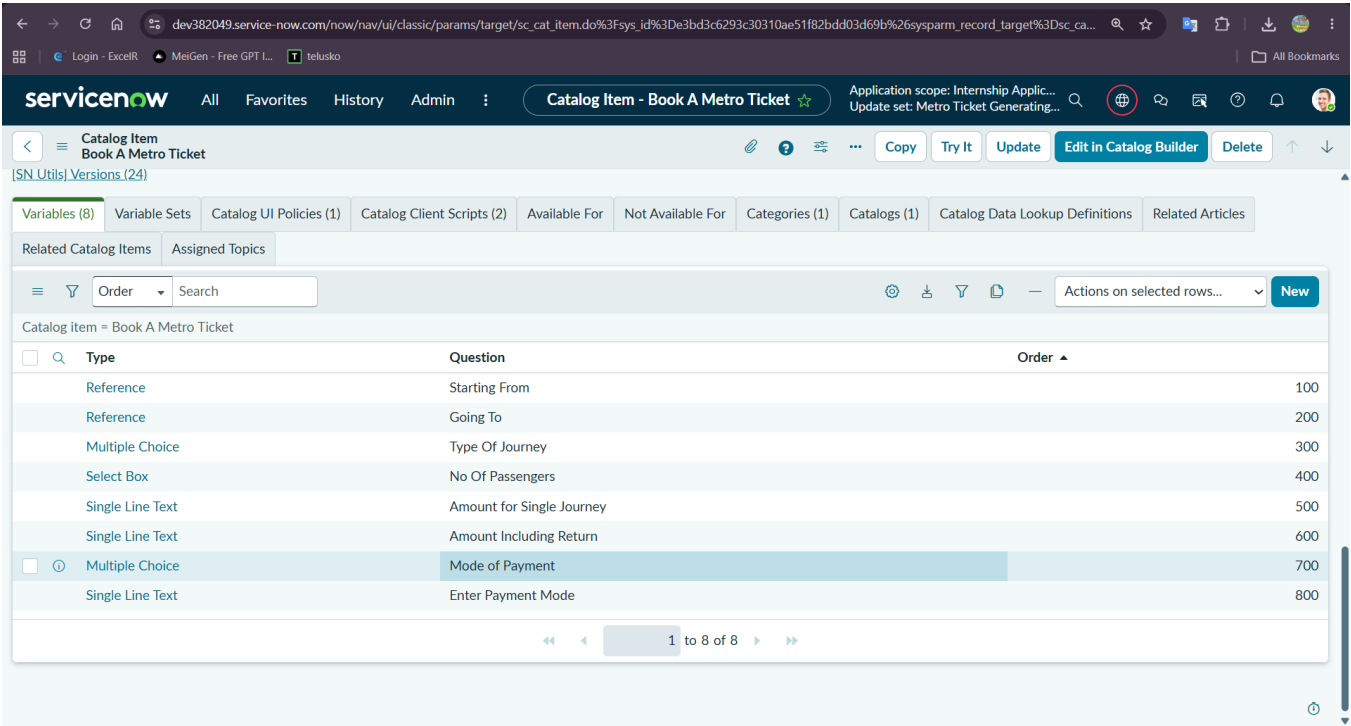
## Final Outcome

Phase 3 transforms the Metro Ticket Generation System from a backend data model into a dynamic, user-friendly ServiceNow booking experience. The catalog item becomes the primary interface through which passengers provide journey details, receive calculated fare values and interact with the QR-ticket experience.

## 21. ServiceNow Implementation Evidence

The following screenshots are appended to the Phase 3 document as implementation evidence. They correspond directly to the catalog item, variable configuration, passenger-facing catalog form, and QR widget activities described in the Phase 3 material.

Figure 21. Catalog Item Variables — Book A Metro Ticket



ServiceNow Maintain Items view showing the eight configured variables: Starting From, Going To, Type of Journey, No Of Passengers, Amount for Single Journey, Amount Including Return, Mode of Payment and Enter Payment Mode.

Figure 22. Catalog Item Configuration — Book A Metro Ticket

dev382049.service-now.com/now/nav/ui/classic/params/target/sc\_cat\_item.do%3Fsys\_id%3De3bd3c6293c30310ae51f82bdd03d69b%26sysparm\_record\_target%3Dsc\_ca...

Login - ExcelRMeiGen - Free GPT 1...telusko

servicenowAllFavoritesHistoryAdminCatalog Item - Book A Metro TicketApplication scope: Internship Applic...Update set: Metro Ticket Generating...

<Catalog ItemBook A Metro TicketCopyTry ItUpdateEdit in Catalog BuilderDelete

Build and modify items faster with the improved [Catalog Builder](#).

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

NameBook A Metro Ticket

CatalogsService Catalog

CategoryServices

State-- None --

Checked out-- None --

OwnerSystem Administrator

Secondary owners

ApplicationInternship Application

Active

Fulfillment automation levelUnspecified

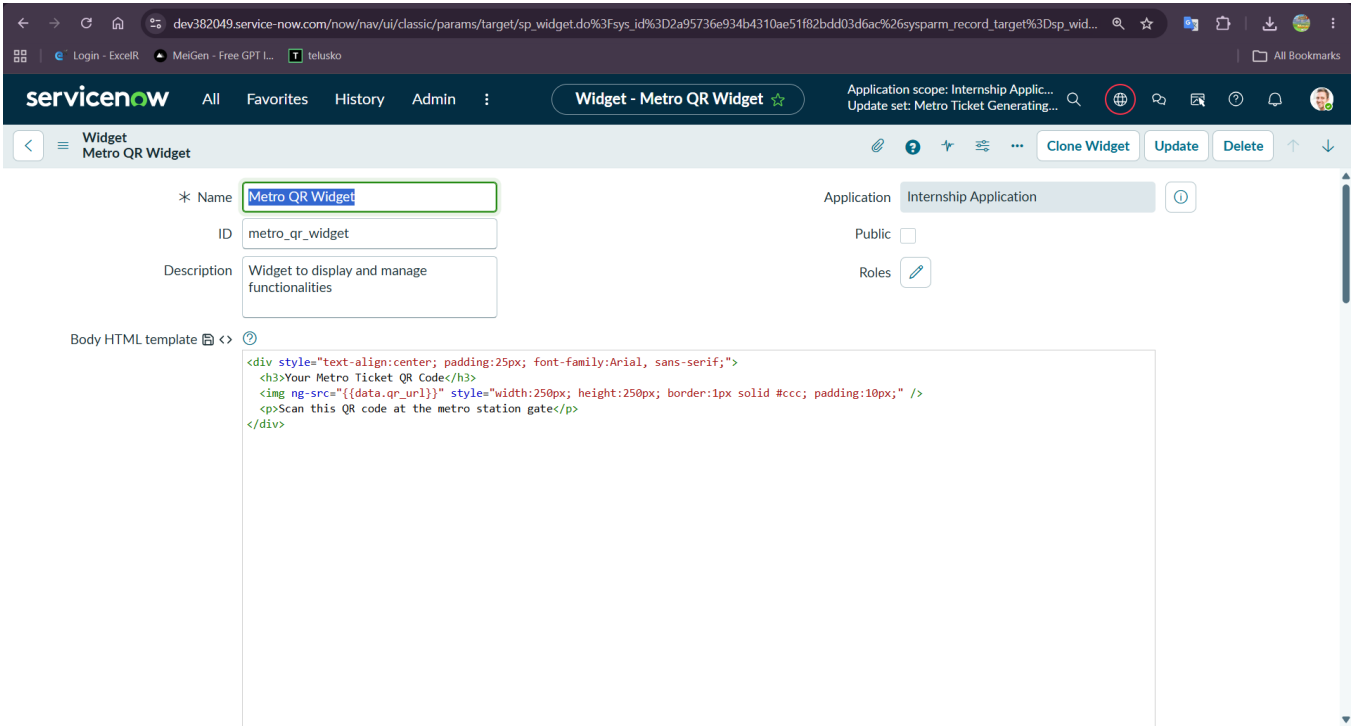
Item DetailsProcess EnginePicturePricingPortal Settings

Short descriptionmetro ticket

Catalog Item form showing the item name, Service Catalog, Services category, active state, application scope and the metro ticket short description.

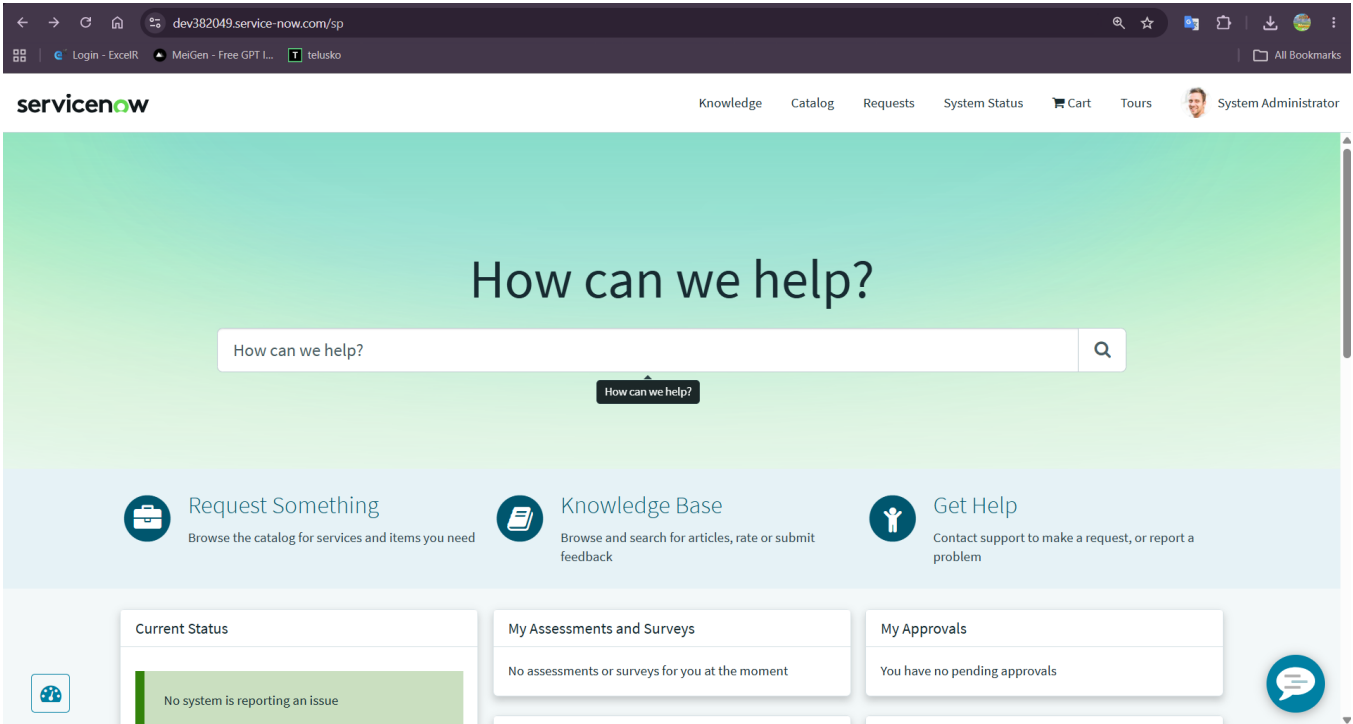
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Figure 23. Service Portal Home — Passenger Entry Point



ServiceNow Service Portal landing page providing the passenger-facing interface and navigation to catalog services.

Figure 24. Book A Metro Ticket — Journey Details



Passenger-facing catalog form showing required Starting From and Going To fields, Single/Return journey selection, passenger count and calculated fare fields.

Figure 25. Book A Metro Ticket — Fare and Payment Section

dev382049.service-now.com/sp?id=sc\_cat\_item&sys\_id=e3bd3c6293c30310ae51f82bdd03d69b

Login - ExcelRMeiGen - Free GPT L...telusko

All Bookmarks

servicenow

KnowledgeCatalogRequestsSystem StatusCartToursSystem Administrator

Home>Service Catalog>Office>Services>Book A Metro Ticket

Search Catalog

Book A Metro Ticket

metro ticket

\* Indicates required

\* Starting From

\* Going To

\* Type Of Journey

☒ Single Journey

☐ Return Journey

\* No Of Passengers

1

Amount for Single Journey

0

Quantity: 1

Delivery Time: 2 Days

Add to Cart

Save as Draft

Order Now

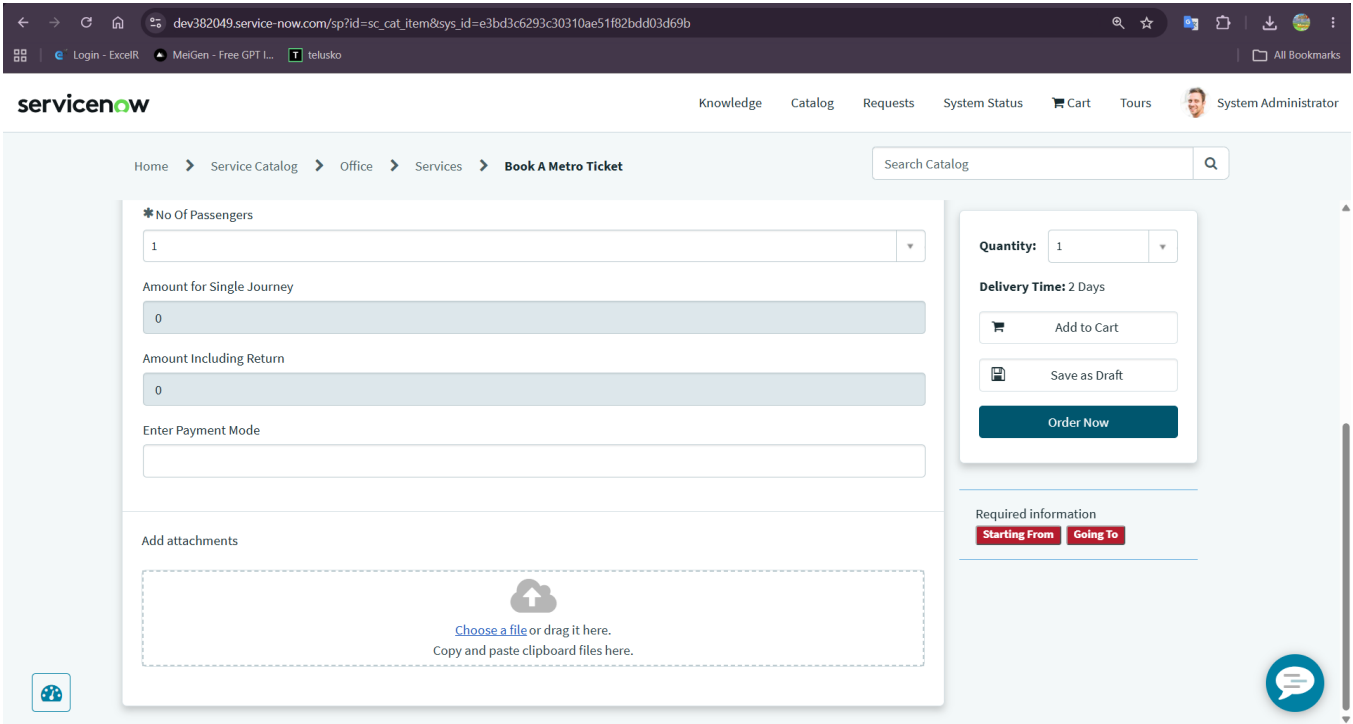
Required information

Starting From

Going To

Lower portion of the catalog form showing passenger count, fare outputs, Enter Payment Mode and checkout controls.

Figure 26. Metro QR Widget Configuration



ServiceNow widget record for Metro QR Widget, including the widget ID and Body HTML template that displays the generated QR image and scan instruction.